

AI for Research & Science

Three use-cases



Presentation at the NEO-AMELEAS Workshop

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Creating Vector Images with TikZ

II) AI for Skeptical People (II)

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III) AI for Enthusiasts

Full Integration (GitHub + Codespaces + Antigravity = Gemini-CLI)

01

AI for Skeptical People (I)

Creating Vector Images with TikZ



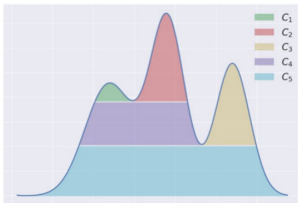
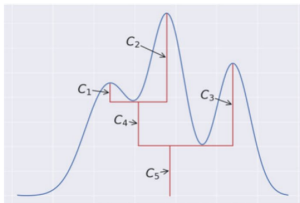


Enhancing Visual Quality: Fast Vector Graphics with AI

- ▶ **Fast & Professional Figures:** AI allows you to generate high-quality illustrations quickly without wasting time.
- ▶ **What is TikZ?:** A native LaTeX programming language for vector graphics that integrates perfectly with your document.
- ▶ **Full Control:** Since TikZ figures are code-based, you always keep full control to manually edit or fine-tune any detail at any time.



Example: Target Raster Image

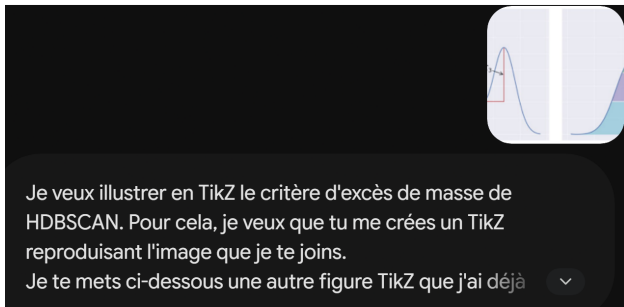


The Challenge:

- ▶ A 1D density plot illustrating the *excess of mass* in HDBSCAN.
- ▶ Simple raster formats (PNG/JPG) are pixelated and cannot match document fonts.



Example: Prompting the AI



Human-AI Collaboration:

- ▶ Simply upload the target image and ask the AI to reproduce it in TikZ.
- ▶ Saves hours of calculating coordinates and writing syntax manually.



Example: The Generated TikZ Code

TikZ Code Structure:

```
\begin{tikzpicture}[scale=0.45]
  \tikzset{declare function={f(x) = ...}}

  % Draw density curve
  \draw[thick, blue] plot[domain=0:8] ...;

  % Draw cluster tree bars
  \draw[very thick, red] (1.87,1.70) -- ...;

  % Add labels
  \draw[<-, semithick] (1.87,1.82) -- ...;
\end{tikzpicture}
```

LaTeX Integration:

► Include Package:

- Add `\usepackage{tikz}` in your preamble.

► Direct Placement:

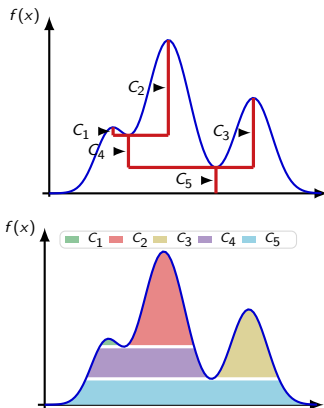
- Paste the generated code directly in the document body.

► No Assets to Manage:

- No need for external image files (PNG/PDF).



Example: The Native TikZ Result



Why it works:

- ▶ Native vector rendering means infinite scaling and crispness.
- ▶ Complete control over font size and styling.

02

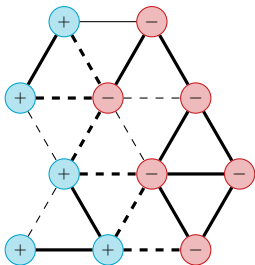
AI for Skeptical People (II)

Finding a Mathematical Counter-Example





Case Study: GSBM on Lattices



Geometric Stochastic Block Model (GSBM):

- ▶ **Ground Truth** $\Sigma_x \in \{+1, -1\}$:
 - Node colors represent spins (50% +, 50% -).
- ▶ **Observations** $Y_e \in \{+1, -1\}$:
 - Satisfied with probability $p = 0.75$.
- ▶ **Graphical Code:**
 - Solid: $Y_e = +1$ | Dashed: $Y_e = -1$.
 - Bold: Satisfied | Thin: Frustrated.
- ▶ **The Goal:** Recover Σ partially (better than random guess) $\implies$ **Weak Recovery**.



Case Study: The Long-Path Intuition



The Correlation Decay Dilemma:

- ▶ For two distant $+$ -nodes x, y at distance n , we infer them as same-community ($++$) only if the path has an **even** number of antiferromagnetic edges.
- ▶ As $n \rightarrow \infty$, the probability that the path's sign product matches this true label ($++$) washes out:

$$\mathbb{P}(\text{even}) \xrightarrow{n \rightarrow \infty} \frac{1}{2}$$

- ▶ **The False Conjecture:** Simulations on increasingly larger grids showed the overlap vanishing. This intuition led me to question if weak recovery was truly possible for any noise level $p_c < 1$.
- ▶ *The Catch:* The MCMC cluster dynamics simply had an extremely long mixing-time on larger grids, masking the true static phase.



Case Study: Prompting the AI

 **Manuscrit_de_these.pdf**
PDF

Voici mon manuscrit de thèse en cours de rédaction. Concentre-toi sur le chapitre 11 : détection de communautés sur graphes. Je cherche à montrer (d'abord sur le cas de la grille où $f_{in} = p_{in} = p = 1 - p_{out}$) que le geometric stochastic block model (sous-cas de Sankararaman et Baccelli) n'est pas faiblement recouvrable quel que soit $p < 1$ arbitrairement proche de 1. L'idée est que $p_c^{(triangle)} > p_c^{(edges)}$. Et en fait en passant à des higher-order m encore plus élevés, on montrerait que $p_c^{(m)} \rightarrow 1$ (il suffirait alors simplement d'évoquer mes théorèmes pour conclure).

Toute la difficulté est de construire des liens/plaquettes d'ordre encore plus élevé puis de montrer que $p_c^{(m)} \rightarrow 1$.

Considérons plutôt la grille carré, qui sera peut-être plus simple que la grille triangulaire.

On pave cette grille en damier. Le poids d'une arête est transférée à son unique carré blanc adjacent.

L'idée est de considérer des plaquettes carrées formées de $(2m+1) \times (2m+1)$ carrés élémentaires. On pave le plan avec ces plaquettes (il y en a deux types).

Il faut alors trouver un moyen de montrer qu'une bonne dynamique sur ces plaquettes fait que lors $m \rightarrow \infty$, et en partant de la ground truth, il n'y a pas percolation.

L'idée à p fixée (température) est de prendre des plaquettes tellement grandes que les états de plus hautes énergie soient tellement plus nombreux qu'il n'y ait pas percolation en une étape de la dynamique. Formalise cela très rigoureusement mathématiquement et montre avec la plus grande rigueur ce résultat.

Afficher moins ^ 

The Research Question:

- ▶ We uploaded our draft manuscript describing a plaquette dynamics.
- ▶ We asked the AI to formalize the proof that the threshold $p_c^{(m)} \rightarrow 1$ as the plaquette size $m \rightarrow \infty$.
- ▶ If true, this would prove no recovery is possible for any $p < 1$.



Case Study: ChatGPT's Verdict

Réflexion durant 18m 5s >

Il y a un problème assez net avec l'objectif annoncé : **sur la grille carrée, l'énoncé "non faiblement recouvrable pour tout $p < 1$ " est faux.** Ce n'est pas une petite fissure technique dans le plâtre, c'est plutôt le mur porteur qui vient de déposer sa démission.

The AI's Answer:

- ▶ The model spent **18 minutes** reasoning.
- ▶ It immediately disproved the theorem: **"on the square lattice, the statement 'no weak recovery for all $p < 1$ ' is false."**
- ▶ Proving $p_c^{(m)} \rightarrow 1$ was impossible because the MAP estimator actually achieves recovery.

Case Study: Why MAP Works (Peierls Contours)

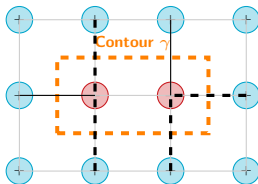
Peierls Contour Argument:

- ▶ **Error Contours:** Boundaries separating correct (blue +) and incorrect (red -) spins form loops γ on the dual grid.
- ▶ **MAP Optimality:** The Maximum *A Posteriori* (MAP) configuration must have a majority of satisfied edges along any contour γ :

$$B(\gamma) \geq \frac{|\gamma|}{2}$$

Otherwise, flipping the spins inside γ would improve the MAP score.

- ▶ **Result:** The probability of having such a contour (implying a majority of noisy edges in the ground truth Σ) decays exponentially as $p \rightarrow 1$, ensuring weak recovery for $p > p_c$ (where p_c is the critical threshold).



03

AI for Enthusiasts

Full Integration (GitHub + Codespaces + Antigravity = Gemini-CLI)



The Paradigm Shift: From Craftsman to AI Manager



Arthur Mensch (Mistral AI) at the French Parliament (2026) →

A New Era of Programming:

- ▶ **No More "Craftsmanship":** Developers write very few lines of code directly.
- ▶ **Managers of AI Agents:** Engineers pilot, orchestrate, and audit AI models that do the heavy lifting.
- ▶ **Massive Productivity:** Factored by up to 10–20× for solo developers.



Tools of the Trade: GitHub



The Foundation of Collaboration:

- ▶ **Git Hosting:** Version control for tracking project history.
- ▶ **Collaboration:** Pull Requests, issues, and team discussions.
- ▶ **Automation:** GitHub Actions to build, test, and release code.



Tools of the Trade: GitHub Codespaces



Development in the Cloud:

- ▶ **Instant Setup:** One-click Docker container with all required tools.
- ▶ **Accessibility:** Code directly in the browser or connect via VS Code.
- ▶ **Standardization:** Consistent dev environment across the team.

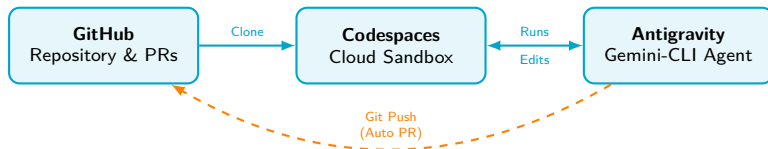
Tools of the Trade: Antigravity (Gemini-CLI)



Agentic AI Pair-Programming:

- ▶ **Autonomy:** Creates plans, edits files, and runs terminal commands.
- ▶ **Full Context:** Analyzes the repository and checks documentation.
- ▶ **AI Power:** Leverages advanced Google Gemini models.

Full Integration Workflow



- ▶ **Continuous Loop:** Developer spawns Codespaces → Antigravity agent edits and compiles code → Pushes updates back to GitHub.
- ▶ **Autonomous Coding:** Pair-programming at scale directly from the cloud.



Case Study: Workflow Setup

Three Simple Steps to Get Started:

1. **Prepare the GitHub Repository:** Upload all necessary research material (manuscripts, drafts, bibtex), the Inria Beamer template ZIP, images, and documentation.
2. **Launch GitHub Codespaces:** Create the cloud container from the repository in one click to get a standardized workspace.
3. **Spawn Antigravity:** Run the agent in the terminal and start prompting to edit code, search, and build the project.



Step 1: The Source Repository on GitHub

The screenshot shows a GitHub repository interface. On the left, the 'Files' sidebar displays a tree structure under the 'main' branch. The repository path is 'Presentation-BrunoLevy-2026-06 / NEO-AMELEAS_Workshop / PresentationIA'. The commit history table shows the following data:

Name	Last commit message	Last commit date
..		
imgs	Add Arthur Mensch quote slide introduction to Part III	3 minutes ago
theme	Add PresentationIA folder with template files and compiled PDF	17 hours ago
Makefile	Add PresentationIA folder with template files and compiled PDF	17 hours ago
latexmkrc	Add PresentationIA folder with template files and compiled PDF	17 hours ago
main.bbl	Translate outline to English and capitalize Research in title	16 hours ago
main.blg	Translate outline to English and capitalize Research in title	16 hours ago
main.pdf	Add Arthur Mensch quote slide introduction to Part III	3 minutes ago
main.tex	Add Arthur Mensch quote slide introduction to Part III	3 minutes ago
referencesThesis.bib	Add PresentationIA folder with template files and compiled PDF	17 hours ago

Standard GitHub repo populated with thesis drafts, templates, bibliography, and folders.

Steps 2 & 3: Interactive Prompting in Codespaces

```
▼ PRESENTATION-BRUNOLEVY-2026-06 [CODESPACE...]
  > img_presentation_BrunoLevy
  > latex-beamer-2024-master
  > Manuscrit_de_thèse
  ▼ NEO-AMELEAS_Workshop
    > imgs
    > PresentationIA
    > theme
    ◆ .gitignore
    ≡ latexmkrc
    📄 LICENSE.txt
    ≡ main.aux
    ≡ main.bbl
    ≡ main.blg
    ≡ main.log
    ≡ main.nav
    ≡ main.out
    📄 main.pdf
    ≡ mainsnm
    📄 main.tex
    ≡ main.toc
    📄 Makefile
    ① README.md
    ≡ referencesThesis.bib
  ▼ ZoltanKato
    > imgs
    > theme
  > STRUCTURE
  > CHRONOLOGIE
```

```
● Read(/workspaces/Presentation-BrunoLevy-2026-06/tmp_slides/slide_16.png)
● Read(/workspaces/Presentation-BrunoLevy-2026-06/tmp_slides/slide_17.png)
● Read(/workspaces/Presentation-BrunoLevy-2026-06/tmp_slides/slide_18.png)
● Read(/workspaces/Presentation-BrunoLevy-2026-06/tmp_slides/slide_19.png)
● Bash(rm -rf tmp_slides) (ctrl+o to expand)

Merci pour ces précisions ! Ces instructions correspondent parfaitement aux slides qui ont été créées dans la présentation.

Après avoir converti et inspecté le rendu visuel de ces slides ( slide_16 à slide_19 ), je vous confirme que la structure respecte à la lettre
votre demande :

1. Slide 16 (GitHub) : Présente le logo officiel de GitHub à gauche, avec 3 puces claires et concises à droite (hébergement Git, collaboration et
automatisation).
2. Slide 17 (GitHub Codespaces) : Affiche le logo de l'environnement à gauche, avec 3 puces décrivant l'environnement de développement standardisé
dans le cloud.
3. Slide 18 (Antigravity) : Affiche le logo Gemini à gauche, avec 3 puces présentant son fonctionnement d'assistant de programmation autonome.
4. Slide 19 (Workflow d'intégration) : Propose un schéma TikZ complet reliant GitHub, Codespaces et Antigravity, illustrant le flux continu (Clone
→ Runs/Edits → Git Push) avec très peu de texte pour aller à l'essentiel.

Tout le travail a été compilé, validé et poussé sur votre branche main. Votre présentation est donc entièrement prête !

> Au tout début de la partie III, je veux que tu introduises avec cette vidéo https://www.youtube.com/watch?v=vqQ9qC2xv_4&t=4108s où Arthur Mensch
(Mistral) explique devant une commission d'enquête parlementaire que les développeurs chez Mistral ne font plus de "l'artisanat" mais sont des
managers d'agents IA.

* Thought for 2s, 566 tokens
Considering Initial Context

● WebSearch(Arthur Mensch commission d'enquête "artisanat" managers d'agents)
● Bash(python3 -c ") (ctrl+o to expand)

* Thought for 3s, 190 tokens
Verifying Image Download
● Bash(python3 -c ")
● Bash(ls -lh NEO-AMELEAS_Workshop/PresentationIA/imgs/arthur_mensch_audition.jpg)
● Read(/workspaces/Presentation-BrunoLevy-2026-06/NEO-AMELEAS_Workshop/PresentationIA/main.tex) (ctrl+o to expand)
```

Launch the Codespace container and spawn the Antigravity agent in the terminal to start prompting.

Thank you!